

M-1 LED Matrix Controller

WLED-MM controller board for HUB75 LED matrix panels with Home Assistant

The Apollo M-1 is a controller board for HUB75 LED matrix panels, designed specifically to drive the HUB75 panel Apollo sells. It runs the open-source WLED-MM (MoonModules) firmware, connects over Wi-Fi, and works with or without Home Assistant for animated displays, status boards, and notifications.

This product is the controller only — the LED matrix (HUB75) panel is not included. Once paired with a panel, the M-1 can run WLED effects and presets, display animated GIFs, and overlay live information such as the time, temperature, humidity, and Home Assistant alerts using multiple WLED segments.

Key Specifications

Product	M-1 LED Matrix Controller (controller only)
Manufacturer	Apollo Automation (Versailles, KY, USA)
Function	Controller board for HUB75 LED matrix panels
Firmware	WLED-MM (MoonModules WLED-MM) — open source
Designed for	Apollo HUB75 LED matrix panel (sold separately)
Microcontroller	ESP32-based
Connectivity	Wi-Fi
Integration	Home Assistant (optional) and standalone WLED
Panel included	No — LED matrix (HUB75) sold separately

Configurations & Pricing

Item	Price (USD)
M-1 LED Matrix Controller (controller only)	\$27.99

The LED matrix (HUB75) panel is not included and is required for use. A full kit with the matrix panel (and optional microphone) is offered separately.

Features & Use Cases

- **WLED effects & presets:** run the full library of WLED-MM animations and presets on a HUB75 panel.
- **Animated GIFs:** display animated GIFs (e.g. 64 x 64) for custom graphics and alerts.
- **Segmented overlays:** use multiple WLED segments to overlay time, temperature, humidity, and messages on an animation.
- **Home Assistant alerts:** trigger displays from HA events — e.g. a door left unlocked, low battery, or poor air quality.
- **Works standalone:** usable with or without Home Assistant via the WLED interface.

Important Notes

- **Controller only:** a HUB75 LED matrix panel is required and not included; the M-1 is designed for Apollo's panel.
- **Open-source firmware:** built on WLED-MM by MoonModules, with thanks to Netmindz; WLED 16.0 / HUB75 support is being evaluated.
- **GIF handling:** resizing and loading GIFs works through WLED but may occasionally need a reset to display — a known firmware quirk.

Documentation & Resources

- **Product page:** apolloautomation.com/products/m-1-led-matrix-controller-only
- **WLED-MM firmware:** github.com/MoonModules/WLED-MM
- **Wiki / documentation:** wiki.apolloautomation.com
- **Community:** [Apollo Automation Discord](#)